

UC-01: Submit Damage Report

Primary Actor – Citizen Reporter

GOAL

Submit a photograph and details of infrastructure damages so that the system can create and route a municipal service ticket.

PRECONDITIONS

- The citizen has access to the reporting application
- The citizen is authenticated if authentication is required
- The citizen has a photograph of the infrastructure damage
- The reporting service is available

POSTCONDITIONS

Success

- A Unique ticket is created.
- The damage report is stored
- The incident location is associated with the report
- The report is routed to the appropriate municipal department
- The citizen receives a submission confirmation

Failure

- No incomplete ticket is created
- The citizen is informed of the missing/invalid information

MAIN SUCCESS SCENARIO

1. Citizen opens the damage reporting function
2. System displays the reporting form
3. Citizen selects the infrastructure damage category
4. Citizen uploads a photograph of the damage
5. System extracts GPS coordinates from the photograph's EXIF metadata
6. System displays the identified location on the municipal GIS map
7. Citizen enters a description of the damage and submits the report.
8. System validates the mandatory report information
9. System creates a unique ticket ID
10. System routes the report to the appropriate municipal engineering department
11. System sends a submission confirmation to the citizen
12. System displays the ticket ID and current status to the citizen.

ALTERNATE FLOW A1: GPS LOCATION UNAVAILABLE

At Step 5:

1. The system attempts to extract GPS coordinates from the uploaded photograph
2. The photograph does not contain valid GPS metadata
3. The system informs the citizen that automatic location detection failed
4. The system requests the citizen to provide the incident location manually
5. The citizen selects the location on the map
6. The system validates the manually supplied location
7. The use case resumes at step 7 of the main success scenario.

If the citizen does not provide a valid location, the report remains unsent and the system displays an appropriate error message.